

Technology Stack

Date	1 July 2026
Team ID	SWTID-2026-9706
Project Name	Voice Based Concept understanding Analyser
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>Streamlit</i>	<i>Interactive web interface</i>
2	Backend / Server-Side	<i>Python</i>	<i>Core application logic</i>
3	Database / Data Storage	<i>Streamlit Session State</i>	<i>Stores transcripts, audio features, and scores in-memory during the active session</i>
4	Cloud / Hosting / Deployment	<i>Streamlit Cloud/Local</i>	<i>easy scalability, cost-effective hosting</i>
5	Version Control & CI/CD	<i>Git & Github</i>	<i>Source code</i>
6	Third-Party APIs / Other Tools	<i>OpenAI Whisper, Sentence-BERT, Librosa, SoundFile, Matplotlib, ReportLab</i>	<i>Speech recognition (Whisper), semantic similarity (Sentence-BERT), audio feature extraction and waveform analysis (Librosa, SoundFile), waveform visualization (Matplotlib), PDF generation (ReportLab). ffmpeg is also required as a system-level dependency for Whisper's audio decoding.</i>